

LLM-Driven Semantic ID for Information Diffusion Prediction

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Abstract

Information diffusion prediction is a fundamental problem for understanding how ideas, opinions, and content propagate across social platforms. Existing models mainly rely on learnable ID-based user embeddings or purely structural features, which largely neglect the rich semantic context behind user behavior. Such representations capture who is connected but not why or how they act, leading to brittle performance in long-tail and sparse-user scenarios. Recent advances in Large Language Models (LLMs) have introduced new opportunities to understand user semantics; however, directly injecting continuous high-dimensional embeddings into diffusion prediction models often causes semantic-behavioral misalignment and overfitting. To overcome these limitations, we propose an LLM-driven Semantic ID framework, termed LSID, which translates LLM-derived semantics into discrete, behaviorally aligned representations through hierarchical residual quantization. This discretization mechanism acts as an information bottleneck, filtering redundant semantic variance while preserving diffusion-relevant signals. LSID decouples structural contrastive pretraining from subsequent LLM-based semantic fusion to bridge the semantic-behavioral gap, compresses the aligned features into hierarchical discrete codes, and adaptively fuses the resulting Semantic IDs into an information diffusion prediction backbone. Experiments on two real-world datasets show that LSID achieves superior prediction and consistently outperforms six state-of-the-art baselines.

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*; • **Information systems** → **Social networks**.

Keywords

Social Network Analysis; Public Opinion; Large Language Models; Information Diffusion Prediction

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1 Introduction

Online social networks have become indispensable platforms for information dissemination, opinion sharing, and social interaction. Consequently, predicting information diffusion trajectories is critical for a variety of applications, including public opinion monitoring, viral marketing, and the mitigation of fake news [14, 17, 27, 30]. Early approaches to this problem are predominantly based on probabilistic models and feature-based machine learning methods [3, 4, 29]. With the advent of deep learning, the dominant methodology [15, 18, 23, 31, 33] has shifted to a multi-stage process. Generally, the process involves three key steps: first, each user is assigned a unique learnable ID embeddings; second, Graph Neural Networks (GNNs) are employed to capture the topology of the social network and generate graph-based user embeddings; and finally, a sequence model, such as a Recurrent Neural Network (RNN) or an attention-based model, is used to model the temporal dynamics of the information cascades and predict their future trajectories.

However, the foundation of this common pipeline is **learnable ID-only embeddings** that are randomly initialized and fundamentally overlook the semantic richness embedded in user profiles, generated content, and interaction histories. This design inevitably abstracts away the intrinsic meaning behind user behaviors. A limitation here lies in the *Semantic-Behavioral Misalignment*. We define the *Semantic Space* as the abstract domain capturing user attributes and latent interests (e.g., what a user posts), whereas the *Behavioral Space* describes users' interaction patterns within the social network (e.g., how a user diffuses information). Although related, these spaces often diverge: users with similar diffusion patterns may exhibit markedly different topical interests, while semantically similar users may play distinct roles in diffusion processes. Existing methods, relying exclusively on structural signals, struggle to bridge this gap, resulting in representations that conflate distinct user intents and fail to capture the nuanced “why” behind diffusion behaviors. Consequently, structurally similar users (e.g., two peripheral nodes) are often mapped to indistinguishable representations, even if one is a credible journalist and the other is a bot spreading rumors.



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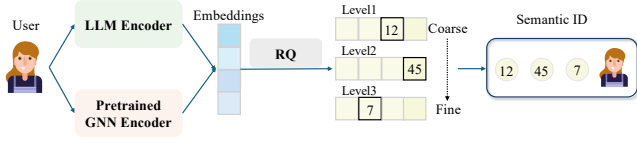


Figure 1: Illustration of Semantic IDs. Instead of using random IDs or raw dense vectors, we compress the multi-view knowledge into a hierarchical, discrete code sequence.

Recent advances in Large Language Models (LLMs) [2, 7] offer a transformative opportunity to reconstruct this missing semantic layer. By decoding user-generated content and inferring latent social roles, LLMs facilitate a paradigm shift from purely structural modeling toward joint structural–semantic diffusion modeling [21, 38]. However, effectively integrating LLM-derived knowledge into diffusion frameworks is non-trivial. Simply concatenating these modalities often yields suboptimal results due to two fundamental challenges: (1) *The Semantic-Behavioral Gap*. Textual semantics and diffusion behaviors reside on distinct, often incongruent manifolds. LLMs are pre-trained to optimize linguistic coherence and topical relevance (e.g., identifying that two users both discuss “politics”), but they are agnostic to the *behavioral dynamics* of propagation (e.g., distinguishing a passive consumer from an active amplifier). Consequently, a naive injection of semantic vectors often leads to *negative transfer*, where the model conflates users who share topical interests but play opposing roles in the diffusion process, thereby destabilizing the structural learning of the backbone predictor. (2) *The Semantic Noise*. Raw embeddings from LLMs (typically 4096-d or larger) are dense and encompass a vast amount of linguistic variation—syntax, style, and general world knowledge, which is largely irrelevant to the specific task of cascade prediction. This high-dimensional redundancy creates a low signal-to-noise ratio for diffusion tasks. Directly utilizing these continuous vectors not only imposes prohibitive storage and computational overheads for large-scale graphs but also exacerbates the risk of overfitting, as the downstream predictor may latch onto spurious linguistic correlations rather than robust, transferable diffusion patterns.

To address these challenges, we propose transforming continuous semantics into compact and composable *Discrete Semantic IDs* (SIDs) for diffusion prediction. The key idea is to impose an *information bottleneck* that filters irrelevant semantic variation while retaining diffusion-relevant user traits. As conceptually illustrated in Figure 1, instead of assigning random atomic IDs, we construct SIDs from *dual-source* knowledge: (i) semantic signals extracted by a frozen LLM from multi-view user prompts, and (ii) structural signals learned by a GNN over the social graph. We instantiate this idea in a framework named **LLM-driven Semantic ID (LSID)** for information diffusion prediction. Concretely, LSID first refines structural embeddings using a multi-view contrastive objective that leverages diffusion-derived behavioral neighborhoods (e.g., temporal co-activation and interaction co-occurrence), then adaptively fuses the refined structure with LLM semantics to form unified user representations. Next, LSID discretizes these representations via hierarchical residual vector quantization, mapping each user to a short code sequence (SID). This discretization induces a coarse-to-fine

organization: users with similar diffusion roles tend to share code prefixes, while later codes capture finer distinctions, enabling compact storage and improving robustness to semantic noise. Finally, LSID retrieves codewords through codebook lookup and injects the reconstructed semantic prior into an attention-based diffusion predictor via gated fusion to predict the next activated user.

The main contributions are summarized as follows:

- We develop a multi-view contrastive refinement mechanism that exploits diffusion-derived behavioral neighborhoods to strengthen structural user representations before semantic fusion.
- We propose a discrete semantic encoding paradigm for diffusion prediction, using hierarchical residual vector quantization to convert continuous multi-view representations into compact and composable Semantic IDs.
- We conduct extensive experiments on two real-world datasets, showing that LSID consistently outperforms six baselines, while providing an interpretable discrete interface for semantic-aware diffusion modeling.

2 Preliminaries

A social network is represented as a graph $G = (U, E)$, where U is the set of users ($|U|=N$) and $E \subseteq U \times U$ is the set of relationships. An edge $(u_i, u_j) \in E$ denotes a connection between user u_i and u_j , such as the friendship or following relationship.

Definition 1. [*Information Diffusion Cascade*] An information diffusion cascade, denoted by C , represents the temporal progression of a piece of information spreading through the network G . Formally, it is defined as a time-ordered sequence of user-activation pairs: $C = \langle (u_1, t_1), (u_2, t_2), \dots, (u_k, t_k) \rangle$ where each tuple (u_i, t_i) signifies that user $u_i \in U$ adopted the information at time t_i . The sequence is strictly ordered by activation time, such that $t_1 < t_2 < \dots < t_k$. The set of all observed cascades is denoted by $\mathcal{C}$.

Definition 2. [*User Profile*] For each user $u \in U$, we define a multi-view profile P_u as a composite representation of their characteristics. Formally, this profile is a tuple: $P_u = (P_u^a, P_u^b, P_u^s)$. The components are: (1) *Attribute Profile* ($P_u^a \in \mathbb{R}^{d_a}$), a vector of static attributes like location and follower count; (2) *Behavior Profile* ($P_u^b \in \mathbb{R}^{d_b}$), a representation of behavior preferences from recent user activity; and (3) *Structural Profile* ($P_u^s \in \mathbb{R}^{d_s}$), a feature vector capturing local topology via metrics like node degree and centrality. The collection of all user profiles is denoted as $\mathcal{P} = \{P_u \mid u \in U\}$.

Problem Statement. The task of *Information Diffusion Prediction* is defined as follows: Given a social network G , a set of user profiles $\mathcal{P}$, and a partial cascade $C_k = \{(u_1, t_1), \dots, (u_k, t_k)\}$ observed up to time t_k , the objective is to learn a predictive function f . This function aims to identify the next user to be activated. Let $V_k = \{u_1, \dots, u_k\}$ be the set of already activated users. For each candidate user $u \in U \setminus V_k$, the function f computes a score indicating the likelihood of activation. The predicted next user, $\hat{u}_{k+1}$, is the one with the maximum score:

$$\hat{u}_{k+1} = \arg \max_{u \in U \setminus V_k} f(u \mid C_k, G, \mathcal{P}) \quad (1)$$

Here, $f(u \mid C_k, G, \mathcal{P})$ is the score assigned to user u , conditioned on the observed cascade, network structure, and user profiles.

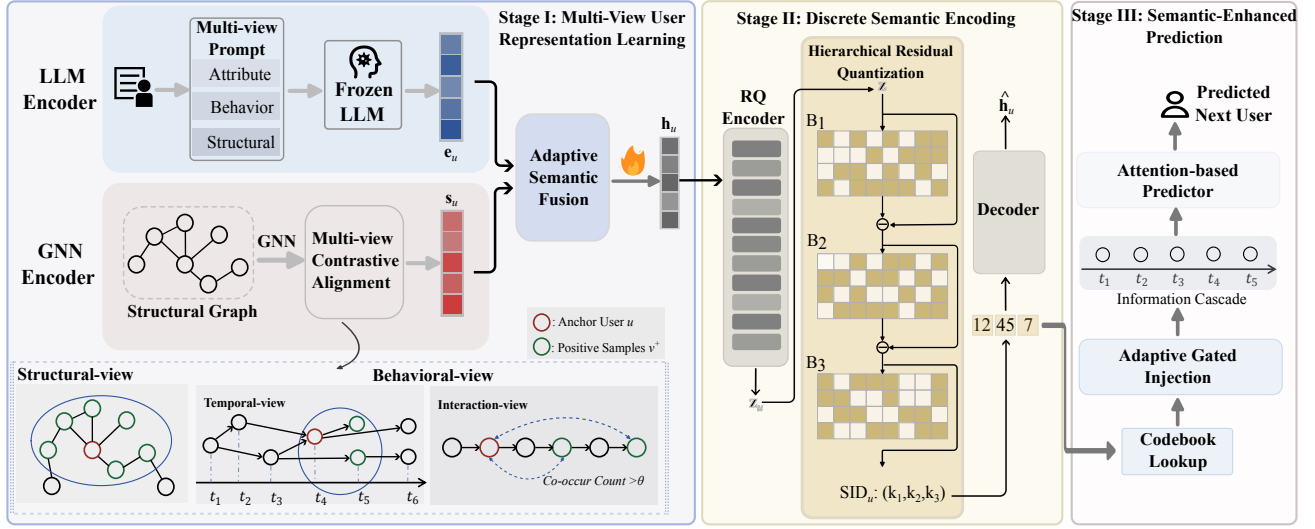


Figure 2: The overview of the proposed LSID framework.

3 Related Work

Information Diffusion Prediction. Information diffusion prediction typically leverages the current cascade information to infer the likelihood of subsequent propagation [8, 31]. Early methods relied on feature engineering and probabilistic models [11, 34], which often required handcrafted assumptions and struggled to generalize across datasets. With the development of deep learning, approaches based on sequential models and GNNs have become mainstream, enabling end-to-end learning from observed diffusion data. One line of research treats diffusion paths as sequences or temporal graphs to model local temporal patterns, while another integrates social networks and diffusion graphs to capture cross-level structural information, such as Topo-LSTM [28], FOREST [32] and Inf-VAE [20]. Building on this foundation, researchers have explored various model architectures to better capture complex diffusion behaviors. For instance, DyHGCN [33], MS-HGAT [23] and MINDS [15] use dynamic heterogeneous graphs and hyper-graph networks to capture user interactions. Some studies also seek breakthroughs through novel representation spaces and learning paradigms: DisenIDP [6] disentangles user intents via hypergraphs, while GODEN [26] and R-ODE [22] adopt Graph ODEs and Ricci curvature to characterize continuous diffusion dynamics. Furthermore, SILN [9] integrates structural and temporal sphere effects, achieving efficient prediction of large-scale social network information diffusion through optimized storage of specific subgraphs in cascades. Despite their structural sophistication, these methods treat users as learnable ID embeddings initialized from random noise. This reliance on structural signals overlooks the intrinsic semantic context of user behaviors, limiting their effectiveness in sparse-user or semantically complex diffusion patterns, where users with similar connectivity may exhibit different behaviors.

LLMs on Social Networks. With the advent of powerful LLMs like GPT [1] and LLaMA [24], social networks are starting to integrate with LLMs, opening new opportunities for incorporating semantic information into user modeling and diffusion analysis. For

example, Social-LLM [13] demonstrates that LLMs can effectively learn user representations at scale by integrating textual profile descriptions with social interaction structures. Recent works also seek to enhance traditional pipelines with LLM-derived knowledge. For example, CARE [38] adopts an in-context learning framework, retrieving historically similar cascades to construct prompts that guide prediction. AutoCas [37] transforms cascade data into textual sequences, leveraging the autoregressive nature of LLMs to forecast popularity. Additionally, MILD [21] introduces an attention-aware framework where LLMs analyze diffusion impacts and user similarities to improve the interpretability of GNN-based predictions. However, these approaches typically employ LLMs as either frozen external feature extractors or independent text generators. They lack a mechanism to align and compress these high-dimensional semantic signals into a compact, diffusion-oriented behavioral space.

4 Methodology

Overview. As illustrated in Figure 2, LSID is a three-stage framework. In Stage I (Multi-View User Representation Learning), we construct a multi-view prompt for each user (attributes, behaviors, and structural descriptors) and extract a semantic embedding e_u from a frozen LLM, while a GNN encoder over the structural graph produces a structural embedding s_u . To make s_u reflect diffusion-relevant relations beyond static topology, we further refine it via a multi-view contrastive objective, and then an adaptive fusion module integrates e_u and the refined s_u into a unified representation h_u . In Stage II (Discrete Semantic Encoding), an RQ-based vector-quantized autoencoder maps h_u to a latent vector z_u , performs hierarchical residual quantization over multiple codebooks to obtain a discrete Semantic ID $SID_u = (k_1, \dots, k_M)$, and reconstructs $\hat{h}_u$ for training. In Stage III (Semantic-Enhanced Prediction), we retrieve codewords indexed by SID_u via codebook lookup, reconstruct an auxiliary semantic feature, and inject it into an attention-based diffusion predictor through adaptive gated injection to infer the next activated user in the cascade.

Prompt	
Role	You are a social data analysis expert, skilled at understanding user characteristics by combining user attribute, social network structures, and content behavior.
Information	<i>Attribute:</i> The user ID is 296. Located in Beijing; has 253 followers and 568 followings; verification status is unverified; the account was created on 2010-06-26. <i>Structural:</i> Has 15 direct neighbors and 9332 second-degree neighbors, with 29 pairs of mutual follow relationships. <i>Behavior:</i> Example of recent content: "Write to each of yourself." "A long-standing rumor".
Question	Based on the above attribute, social network structure, and content behavior, please analyze this user's social role and interest tendencies.

Figure 3: Illustration of the prompt template.

4.1 Multi-View User Representation Learning

Our framework begins with a Multi-View Representation Learning module designed to unify *intrinsic semantics* from user profiles with *extrinsic interaction patterns* from the social graph. By projecting LLM-derived features and graph-based structural signals into a shared latent space, we establish a comprehensive user profile mechanism. This multi-view alignment ensures that the resulting embeddings are not merely structural proxies, but semantically distinguishable representations within the hidden space. Consequently, the model can differentiate between structurally proximate users who exhibit divergent behavioral intents—a granularity often lost in traditional GNN-based methods that rely solely on randomized and independent ID-based initializations.

4.1.1 Prompt Generation and Feature Extraction. While LLMs excel at modeling natural language, they struggle to directly encode structured or numerical social signals (e.g., user IDs and topology-derived metrics). To address this modality mismatch, we construct a multi-view prompt template comprising three components, following prompt-design practices [5, 12, 21, 35]: (1) Role-Playing, (2) User Information, and (3) Task Instruction. Specifically, the *User Information* component serializes the multi-view user profile $P_u = (P_u^a, P_u^b, P_u^s)$ into a textual description, converting attributes, behaviors, and structural descriptors into LLM-compatible natural language. An example prompt is illustrated in Figure 3.

Given the constructed prompt TP_u , we feed it into a frozen, pre-trained LLM and extract the last-layer hidden state of the final valid token. To align the dimension of the semantic space with the structural space, we employ a linear projection layer. The generation of the initial semantic embedding $\mathbf{e}_u$ is formalized as:

$$\mathbf{e}_u = \mathbf{W}_P \cdot \text{LLM}(TP_u), \quad (2)$$

where $\text{LLM}(\cdot) \in \mathbb{R}^{d_{\text{lim}}}$ denotes the raw output from the LLM hidden layer, and $\mathbf{W}_P \in \mathbb{R}^{d \times d_{\text{lim}}}$ are the learnable projection parameters. This yields the semantic matrix $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_N]^\top$.

4.1.2 Multi-View Contrastive Alignment. Parallel to the semantic branch, we employ a Graph Attention Network (GAT) [25], initialized with learnable node embeddings, to generate the structural representation matrix $\mathbf{S}$. However, relying exclusively on static topological connectivity fails to capture the dynamic context of information propagation. To address this, we introduce a multi-view contrastive alignment strategy that refines $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_N]^\top$ by aligning it with diffusion interaction patterns. This approach optimizes the structural embedding via two complementary views: a *Structural View* and a *Behavioral View*, with the latter explicitly characterizing temporal dynamics and interaction strength.

For an anchor user u , we construct a positive sample set $\mathcal{S}_u^+$ by jointly aggregating samples identified from the following criteria:

(i) Structural View. To capture structural proximity, we traverse the network topology to identify neighbors within two hops. We compute a structural importance score for each neighbor based on degree centrality and hop distance. The top- K neighbors with the highest scores are included in $\mathcal{S}_u^+$, ensuring that the learned embedding encodes high-order connectivity patterns.

(ii) Behavioral View. We further decompose behavioral signals into temporal and interaction dimensions to augment $\mathcal{S}_u^+$: (1) *Temporal Correlation*: Analyzing cascade sequences, we identify users who frequently appear within a narrow temporal window relative to the anchor user, reflecting synchronized activation dynamics. (2) *Interaction Patterns*: To model the strength of behavioral ties, we calculate user co-occurrence frequencies within identical propagation chains. Users whose co-occurrence counts exceed a predefined threshold are selected as positive samples, capturing implicit behavioral associations. Note that we construct behavioral statistics using only training cascades to avoid test leakage.

We employ the InfoNCE loss to pull the anchor user u closer to its multi-perspective positive samples $v^+ \in \mathcal{S}_u^+$, while pushing it away from randomly sampled negative users $v^- \in \mathcal{S}_u^-$. The objective function for optimizing $\mathbf{S}$ is defined as:

$$\mathcal{L}_{\text{CL}} = - \sum_{u \in \mathcal{U}} \frac{1}{|\mathcal{S}_u^+|} \sum_{v^+ \in \mathcal{S}_u^+} \log \frac{\exp(\text{sim}(\mathbf{s}_u, \mathbf{s}_{v^+})/\tau)}{\sum_{k \in \mathcal{S}_u^- \cup \{v^+\}} \exp(\text{sim}(\mathbf{s}_u, \mathbf{s}_k)/\tau)}, \quad (3)$$

where $\mathbf{s}_u$ denotes the structural embedding of user u , $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and τ is the temperature parameter. Through this alignment mechanism, the model jointly encodes static structural relations and fine-grained behavioral dynamics.

4.1.3 Adaptive Semantic Fusion. We propose an Adaptive Semantic Fusion mechanism to integrate the semantic knowledge $\mathbf{e}_u$ with the optimized structural signals $\mathbf{s}_u$. We adopt a progressive training strategy where we freeze the structural parameters after the contrastive alignment and exclusively optimize a lightweight fusion network. This network dynamically calibrates the fusion ratio for each feature dimension. The calculation of the gating vector α_u and the final unified representation $\mathbf{h}_u$ is defined as:

$$\alpha_u = \sigma(\mathbf{W}_f [\mathbf{e}_u \| \mathbf{s}_u] + \mathbf{b}_f), \quad \mathbf{h}_u = \text{LN}(\alpha_u \odot \mathbf{s}_u + (1 - \alpha_u) \odot \mathbf{e}_u), \quad (4)$$

where $[\cdot \| \cdot]$ denotes concatenation, σ is the Sigmoid function, $\mathbf{W}_f \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_f \in \mathbb{R}^d$ are learnable parameters, and $\text{LN}(\cdot)$ denotes Layer Normalization. This gating mechanism allows the model to selectively filter noise, producing a robust user embedding that is both structurally distinct and semantically expressive.

4.2 Discrete Semantic Encoding

Following the adaptive semantic fusion stage, we obtain an enhanced user representation matrix $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]^\top \in \mathbb{R}^{N \times d}$, which integrates multi-view structural signals into the semantic space. This part introduces a Discrete Semantic Encoding module based on Residual Quantized (RQ) [16, 19]. By compressing continuous vectors into hierarchical SIDs, we enforce a coarse-to-fine semantic structure where users with similar behaviors explicitly share underlying codewords. This discrete representation addresses the limitations of dense embeddings—such as noise susceptibility and storage overhead—by explicitly capturing shared behavioral patterns through a compact, compositional structure for more robust diffusion modeling.

4.2.1 Hierarchical Residual Quantization. We implement the discretization via a multi-stage residual framework. The core idea is to approximate the continuous vector $\mathbf{h}_u$ as an ordered composition of codewords drawn from a set of learnable codebooks $\mathcal{B} = \{\mathbf{B}_1, \dots, \mathbf{B}_M\}$. Here, M denotes the quantization depth, and each codebook $\mathbf{B}_m = \{\mathbf{q}_1^{(m)}, \dots, \mathbf{q}_K^{(m)}\} \in \mathbb{R}^{K \times d}$ contains K learnable prototypes. To ensure a stable optimization landscape and mitigate potential codebook collapse, we initialize these vectors via the K-means clustering approach.

The encoding process proceeds recursively. First, a dedicated MLP-based encoder projects the fused user embedding $\mathbf{h}_u$ into a latent query vector:

$$\mathbf{z}_u = \text{Encoder}(\mathbf{h}_u). \quad (5)$$

Let $\mathbf{r}_0 = \mathbf{z}_u$ denote the initial residual. In each quantization step m (ranging from 1 to M), the module identifies the nearest codeword in $\mathbf{B}_m$ that best approximates the current residual $\mathbf{r}_{m-1}$. The discrete index k_m is selected by:

$$k_m = \arg \min_{k \in \{1, \dots, K\}} \left\| \mathbf{r}_{m-1} - \mathbf{q}_k^{(m)} \right\|_2^2. \quad (6)$$

Once the optimal codeword $\mathbf{q}_{k_m}^{(m)}$ is retrieved, we update the residual to isolate the unexplained variance for the subsequent quantization layer: $\mathbf{r}_m = \mathbf{r}_{m-1} - \mathbf{q}_{k_m}^{(m)}$. This chain of operations effectively yields a quantized approximation $\hat{\mathbf{z}}_u$, defined as the summation of the selected codeword sequence:

$$\hat{\mathbf{z}}_u = \sum_{m=1}^M \mathbf{q}_{k_m}^{(m)}. \quad (7)$$

To ensure that these discrete codes retain the original semantic fidelity, $\hat{\mathbf{z}}_u$ is passed through an MLP-based decoder to reconstruct the initial user embedding, denoted as:

$$\hat{\mathbf{h}}_u = \text{Decoder}(\hat{\mathbf{z}}_u), \quad (8)$$

where the network is jointly optimized by minimizing the reconstruction error $\|\mathbf{h}_u - \hat{\mathbf{h}}_u\|_2^2$ and a commitment loss to stabilize the codebook updates. The final Semantic ID for user u is defined as the sequence of selected indices $\text{SID}_u = [k_1, k_2, \dots, k_M]$, with a unique instance-specific suffix appended if necessary to resolve semantic collisions and ensure a one-to-one mapping.

4.2.2 Optimization Objective. To ensure that the generated SIDs are semantically accurate and structurally informative, we optimize the RQ module using a composite objective function. This function comprises three components: reconstruction alignment, hierarchical quantization constraints, and codebook diversity regularization.

Reconstruction Loss. The primary goal is to minimize the information loss incurred by the discrete bottleneck, which quantifies the Euclidean distance between the input user embedding $\mathbf{h}_u$ and the reconstructed $\hat{\mathbf{h}}_u$:

$$\mathcal{L}_{\text{rec}} = \sum_{u=1}^N \|\mathbf{h}_u - \hat{\mathbf{h}}_u\|_2^2. \quad (9)$$

Quantization Loss. Since the discrete argmin operation in vector quantization is non-differentiable, we employ a standard quantization loss to update the codebooks and constrain the encoder output. This loss minimizes the discrepancy between the residuals and the selected codewords. It consists of two terms: (1) a *codebook loss*, which moves codewords towards the encoder outputs, and (2) a *commitment loss*, which prevents the encoder outputs from fluctuating excessively:

$$\mathcal{L}_{\text{rq}} = \sum_{u=1}^N \sum_{m=1}^M \left(\|\text{sg}[\mathbf{r}_{m-1}] - \mathbf{q}_{k_m}^{(m)}\|_2^2 + \beta \|\mathbf{r}_{m-1} - \text{sg}[\mathbf{q}_{k_m}^{(m)}]\|_2^2 \right), \quad (10)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator, and β is a hyperparameter balancing the commitment constraint.

Diversity Regularization. To mitigate codebook collapse (i.e., only a fraction of codewords are utilized), we introduce an auxiliary diversity loss to encourage uniform codeword distribution per batch. We minimize the usage variance to ensure all codewords are utilized equally. Let $n_k^{(m)}$ denote the number of times the k -th codeword in layer m is selected within the current batch, and $\bar{n}^{(m)}$ be the average usage count. The loss is defined as:

$$\mathcal{L}_{\text{uni}} = \sum_{m=1}^M \frac{1}{K} \sum_{k=1}^K \left(n_k^{(m)} - \bar{n}^{(m)} \right)^2. \quad (11)$$

This ensures that the discrete space is fully exploited, enhancing the discriminative power of the SIDs.

Total Objective. The final training objective is a weighted combination of the aforementioned components:

$$\mathcal{L}_{\text{SID}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{rq}} + \lambda \mathcal{L}_{\text{uni}}, \quad (12)$$

where λ controls the strength of the diversity regularization.

The resulting SID_u provides a structured, interpretable format for user profiling. For instance, with codebook depth $M = 3$, a user might be mapped to $\text{SID}_u = \langle 12, 45, 7 \rangle$, where the first index (12) captures the dominant global feature (e.g., broad interest domain), and subsequent indices (45, 7) recursively encode residual information, adding fine-grained attributes (e.g., specific behavioral patterns). This structure induces a *prefix-sharing mechanism*: a neighbor user encoded as $\langle 12, 45, 9 \rangle$ shares the prefix $\langle 12, 45, \cdot \rangle$ with the anchor user, allowing the downstream diffusion model to capture neighborhood similarity through shared token segments. Section 5.5 provides visualization and analysis of these hierarchical clusters to validate this semantic alignment.

4.3 Semantic-Enhanced Prediction

Leveraging the compactness of the SIDs constructed in the previous stage, we propose an effective injection mechanism to enhance downstream cascade prediction. Rather than storing the memory-intensive dense vectors, we retain only the discrete SID_u for each user. This significantly minimizes storage overhead while allowing for the effective retrieval of semantic priors.

4.3.1 Adaptive Gated Injection. We adopt the state-of-the-art predictor SILN [9] as our backbone. During the inference phase, we dynamically reconstruct the quantized semantic vector $\hat{z}_u$ *on-the-fly* by retrieving and summing the corresponding codewords from the frozen codebooks, augmented by the suffix identifier embedding.

To align this pre-trained semantic signal with the diffusion backbone, we project it into the target feature space via a linear transformation, yielding the auxiliary semantic feature $\mathbf{h}_{\text{sem}} = \mathbf{W}_{\text{proj}}\hat{z}_u + \mathbf{b}_{\text{proj}}$. Simultaneously, the backbone generates a task-specific user representation $\mathbf{x}_u$, which is learned via GNNs. To effectively fuse these modalities, we employ a dimensional gating mechanism that models the non-linear interactions between the structural backbone and the semantic prior:

$$\mathbf{g}_u = \sigma(\mathbf{W}_g[\mathbf{x}_u \parallel \mathbf{h}_{\text{sem}} \parallel (\mathbf{x}_u - \mathbf{h}_{\text{sem}}) \parallel (\mathbf{x}_u \odot \mathbf{h}_{\text{sem}})] + \mathbf{b}_g), \quad (13)$$

where σ is the sigmoid activation, $[\cdot \parallel \cdot]$ denotes concatenation, and $\odot$ represents element-wise multiplication. The explicit difference and product terms are included to capture the discrepancy and alignment between the two views. The final enhanced representation $\mathbf{h}_u^{\text{final}}$ is obtained via:

$$\mathbf{h}_u^{\text{final}} = (1 - \mathbf{g}_u) \odot \mathbf{x}_u + \mathbf{g}_u \odot \mathbf{h}_{\text{sem}}. \quad (14)$$

This gating mechanism allows the model to selectively refine the backbone embedding $\mathbf{x}_u$ using the robust semantic knowledge from $\mathbf{h}_{\text{sem}}$, filtering out irrelevant noise.

4.3.2 Diffusion Prediction. Given a partial cascade sequence C_k , the sequence of enhanced representations $\mathbf{H}_C = [\mathbf{h}_{u_1}^{\text{final}}, \dots, \mathbf{h}_{u_k}^{\text{final}}]$ is fed into the backbone’s Temporal and Structural Attention modules to generate a dynamic context vector $\mathbf{o}_t$ at each time step t_k .

We calculate the probability distribution $\hat{\mathbf{y}}_{k+1} \in \mathbb{R}^{|U|}$ for the next active user. To ensure validity, a mask vector $\mathbf{M}$ is applied to exclude users who have already appeared in the history:

$$\hat{\mathbf{y}}_{k+1} = \text{Softmax}(\mathbf{W}_o \mathbf{o}_t + \mathbf{M}), \quad (15)$$

where $\mathbf{W}_o$ projects the context into the user space.

4.3.3 Training Objective. We optimize the prediction backbone while keeping the SIDs and codebooks fixed. The model minimizes the cross-entropy loss between the predicted probability and the ground-truth next user. The objective function for a set of training cascades C_{train} is formulated as:

$$\mathcal{L}_{\text{task}} = - \sum_{C \in C_{\text{train}}} \sum_{k=1}^{|C|-1} \log(\hat{y}_{k+1, u_{k+1}}), \quad (16)$$

where $\hat{y}_{k+1, u_{k+1}}$ denotes the predicted probability of the true next user u_{k+1} after the current step t_k . In this way, the learnable parameters of the backbone and the projection layers are updated.

Table 1: Statistics of the News and Weibo Datasets

Dataset	# Cascades	Avg. Len.	# Users	# Links
News	1,291 + 180	67.88	10,254	83,959
Weibo	1,910 + 522	183.64	31,061	294,755

4.4 Complexity Analysis

The computational cost of our framework mainly arises from GNN-based aggregation $O((N + |E|)d^2)$ and RQ quantization $O(N \cdot M \cdot K \cdot d)$ for constructing user representations, where N and $|E|$ denote the number of users and edges, and M and K represent the codebook depth and size. Since LLM inference is performed as a one-time preprocessing step, its cost is excluded from the training complexity. During inference, semantic vector composition via codebook lookup incurs $O(L \cdot M \cdot d)$, which is effectively constant with respect to the network size. The adaptive gating injection involves linear projections, resulting in $O(L \cdot d^2)$ complexity for a cascade sequence of length L . These costs are negligible compared to the $O(L^2 \cdot d)$ complexity of the backbone’s self-attention mechanism, ensuring that LSID maintains high efficiency for forecasting.

5 Experiments

5.1 Experimental Settings

5.1.1 Dataset Description. We conduct experiments on two real-world datasets, Weibo and News, both of which provide a social network structure and associated information cascades [21, 36]. Each dataset is released in two parts: a *train* split that contains cascades and the underlying graph signals, and a *raw* split that additionally includes rich textual information (e.g., user profiles and posts). To prevent temporal leakage, we sort the *raw* split chronologically and perform a time-aware partition. Specifically, the final training set consists of the entire *train* split plus the earliest 50% of *raw*. The remaining 50% of *raw* is divided into validation and test sets (25% each), both drawn from the later time period. Table 1 provides the statistics, where #Cascades indicates #train + #raw.

5.1.2 Evaluation Metrics. Following prior studies [9, 15, 23], we use Hits rate (Hits@K) and Mean Average Precision (MAP@K) to evaluate the performance on benchmark datasets. For a comprehensive evaluation, $k \in \{10, 50, 100\}$.

5.1.3 Baselines. We compare our method against the following six methods: (1) **DyHGCN** [33] jointly learns user representations from social structures and diffusion history using dynamic heterogeneous GCNs. (2) **MSHGAT** [23] models high-order diffusion relationships with sequential hypergraphs and applies self-attention to capture cascade dynamics. (3) **DisenIDP** [6] disentangles user representations via intent-aware hypergraphs and models long- and short-term temporal dependencies. (4) **GODEN** [26] employs ODE-based GNNs to model continuous-time user dynamics and evolving preferences. (5) **CARE** [38] adopts an in-context learning framework by retrieving relevant historical cascades to enhance current predictions. (6) **SILN** [9] introduces structural and temporal modules to model differentiated participant influences.

Table 2: Hits@K and MAP@K on two datasets (%), scores are the higher the better.

Models	News						Weibo					
	H@10	H@50	H@100	M@10	M@50	M@100	H@10	H@50	H@100	M@10	M@50	M@100
DyHGCN	16.93	24.32	28.02	11.13	11.48	11.53	45.09	59.36	64.83	28.55	29.25	29.34
MSHGAT	17.09	25.81	30.42	10.20	10.58	10.66	39.47	60.06	67.84	21.22	22.20	22.31
DisenIDP	11.87	26.07	33.01	5.01	5.63	5.73	38.39	62.60	68.30	21.48	22.42	22.50
GODEN	21.78	33.33	37.86	13.41	13.94	14.04	48.79	65.14	71.56	30.09	30.90	30.99
CARE	20.45	32.14	36.92	11.60	12.31	12.40	47.31	62.89	69.30	27.60	28.85	28.95
SILN	26.40	33.85	37.31	15.35	15.72	15.77	58.46	68.84	72.28	38.94	39.47	39.52
LSID	29.03	39.45	44.62	16.92	17.44	17.51	62.29	74.59	78.39	40.89	41.52	41.58

5.1.4 Implementation Details. The experiments are conducted on NVIDIA RTX 4090 (24GB). We adopt ChatGLM3-6B [10] as the LLM backbone, which is standardized into 128-dimensional user semantic vectors. For contrastive learning, positive samples are defined by 2-hop neighbors within a 2-hour window, with interaction thresholds set to $\theta = 10$ for News and $\theta = 20$ for Weibo. In the discrete encoding phase, user embeddings are quantized into 3 and 4 sub-codes with codebook sizes of 256 and 64 for the News and Weibo datasets, respectively. The weight of diversity loss is set to $\lambda = 0.25$. For the SILN [9] backbone with semantic injection, we set attention heads to 16 for News and 24 for Weibo, with dimension 256. The model is optimized using Adam with a learning rate of 10^{-3} and a dropout ratio of 0.3. All baselines follow their original configurations and are reimplemented for fair comparison.

5.2 Quality Evaluation

Table 2 presents the overall performance comparison. Specifically, LSID outperforms the strongest baseline (SILN) by substantial margins: on the News dataset, it relatively improves Hits@100 by 7.31% and MAP@100 by 1.74%; on Weibo, it achieves a 6.11% gain in Hits@100. The consistent superiority across different cutoffs (K=10, 50, 100) indicates that LSID enhances overall ranking quality rather than merely optimizing shallow recall. While these recent methods improve upon traditional topology-based models by capturing cascade dynamics, they remain limited by the absence of semantic information. In contrast, LSID consistently achieves state-of-the-art performance across all metrics on both datasets. These gains are attributed to three key mechanisms: (1) Semantic Alignment, which effectively bridges the gap between heterogeneous LLM-derived semantics and GNN-based structural features; (2) Discrete Encoding, where the discrete RQ bottleneck filters out irrelevant linguistic variance, clustering users by latent interests; and (3) Adaptive Injection, which allows the backbone to selectively incorporate external semantics without disrupting original diffusion patterns.

5.3 Ablation Study

5.3.1 User Representation Learning Module. In Table 3, we obtain the **w/o LLM** variant by deriving user representations solely from GNN-based structural embeddings, and the **w/o CL** variant by directly fusing structural and semantic features without contrastive learning. The decline in **w/o LLM** suggests that structural features alone lack context-aware semantic information. Meanwhile, the

Table 3: Ablation study of key components.

Models	News		Weibo	
	Hits@100	MAP@100	Hits@100	MAP@100
w/o CL	42.42	16.91	78.07	41.02
w/o LLM	43.05	16.82	77.68	40.91
w/o RQ	42.13	16.54	76.67	40.58
w/o $\mathcal{L}_{rq}$	43.25	17.30	78.04	41.06
LSID	44.62	17.51	78.39	41.58

Table 4: Comparison with different LLMs on News dataset.

Metric	ChatGLM3-6B	LLaMA3-8B	Qwen1.5-7B
Hits@100	44.62	44.39	44.53
MAP@100	17.51	17.35	17.42

lower performance of **w/o CL** highlights that a naive fusion of heterogeneous features is suboptimal. This confirms the necessity of both the semantic source and the alignment mechanism.

5.3.2 Discrete Encoding Module. We observe performance degradation when removing the quantization loss constraint (**w/o $\mathcal{L}_{rq}$**) or replacing learned SIDs with random IDs (**w/o RQ**). Notably, the worst performance is observed with **w/o RQ**, which verifies that the improvement stems from leveraging content-based semantic information encoded in the hierarchical structure, not the discrete format itself. The lower results of **w/o $\mathcal{L}_{rq}$** also validate the role of codebook regularization in ensuring a stable latent space approximation during model optimization.

5.4 Sensitivity Analysis

5.4.1 Impact of LLM Backbones. A key feature of our LSID framework is that the LLM acts as a pluggable semantic encoder, decoupled from the downstream model, making the framework independent of any specific language model architecture. To examine the robustness of LSID under different semantic backbones, we evaluate its performance on the News dataset using three representative LLMs with similar parameter scales: ChatGLM3-6B [10], LLaMA3-8B [7], and Qwen1.5-7B [2]. We use the same prompt template and extraction rule for all backbones and keep the LLM frozen. As

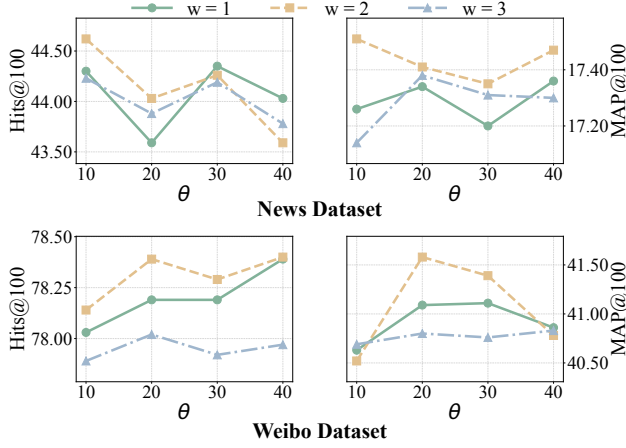


Figure 4: Hyperparameters time window (w) and interaction threshold (θ) on News and Weibo datasets.

shown in Table 4, LSID exhibits stable performance across different backbones. This robustness stems from a multi-view contrastive learning and adaptive fusion strategy that aligns semantic and structural embeddings into a unified diffusion-aware representation, demonstrating the model independence of our framework.

5.4.2 Impact of Key Hyperparameters. We examine the impact of four key hyperparameters: Time Window (w), Interaction Threshold (θ) for contrastive learning, and Quantization Depth (M) and Codebook Size (K) for discrete encoding. The results are illustrated in Figure 4 and Figure 5. **(1) Impact of w and θ .** We observe a trade-off between information gain and noise. For the Time Window, $w = 2$ consistently yields optimal results, balancing context capture against temporal noise. Regarding Interaction Threshold, News peaks at a lower value ($\theta = 10$). In contrast, Weibo favors a higher threshold ($\theta = 20$), indicating a need for stricter filtering to remove interaction noise. **(2) Impact of M and K .** Results indicate distinct preferences for granularity. News achieves best results with $M = 3$, $K = 256$, favoring a larger vocabulary for distinct topics. In contrast, Weibo prefers $M = 4$, $K = 64$, requiring deeper composition for complex behaviors. Overall, performance degrades with insufficient capacity or excessive redundancy, emphasizing the need for balanced compactness to maintain effective representation.

5.5 Case Study

To evaluate semantic expressiveness, we quantify the relationship between SIDs matching depth and topic consistency by mapping users' participations to predefined topic categories, which are derived via keyword clustering on users' historical posts, and then computing the Jaccard similarity across 20,000 randomly sampled Weibo user pairs. Pairs are grouped into Low ($s < 0.15$), Mid, and High ($s \geq 0.5$) consistency levels. Figure 6 reveals a strong positive correlation: pairs with disjoint SIDs exhibit low similarity, while the proportion of High-consistency pairs rises sharply at Level 3. This confirms that our hierarchical quantization effectively preserves semantic structure, where shared prefixes capture coarse-grained interests and deeper matches reflect fine-grained homogeneity.

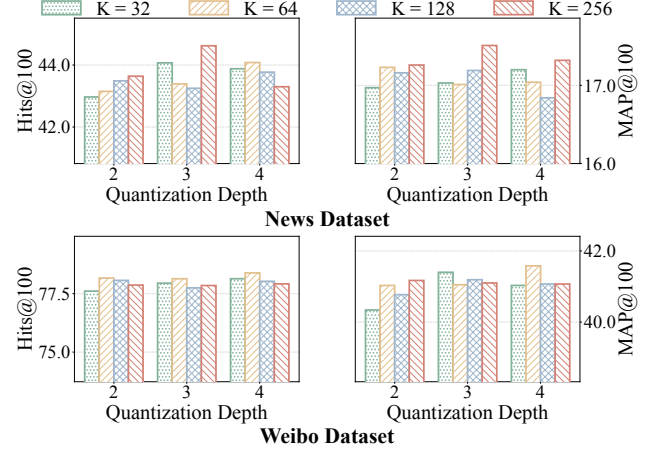


Figure 5: Hyperparameters quantization depth (M) and codebook size (K) on News and Weibo datasets.

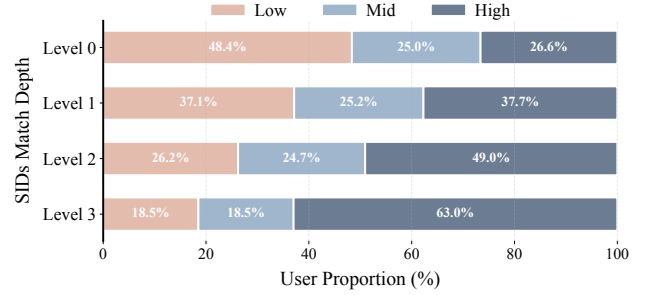


Figure 6: Case study of semantic alignment quality.

6 Conclusion

In this paper, we introduce LSID, a semantic-enhanced framework that bridges the gap between static structural signals and dynamic user intents in information diffusion prediction. By synthesizing LLM-derived semantics with diffusion behaviors, LSID constructs discrete, hierarchical Semantic IDs (SIDs) via a residual quantization mechanism. This approach not only compresses high-dimensional user traits into compact, storage-efficient codes but also facilitates the “coarse-to-fine” alignment of structural neighbors with semantic cohorts. Extensive experiments on real-world datasets demonstrate that LSID significantly outperforms state-of-the-art baselines, proving the effectiveness of injecting discrete semantic priors into diffusion backbones. In future, we plan to explore autoregressive transformers to model the probability distribution of SIDs directly. This would enable the generation of synthetic, yet realistic, diffusion cascades, facilitating counterfactual analysis and public opinion simulation without compromising user privacy.

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